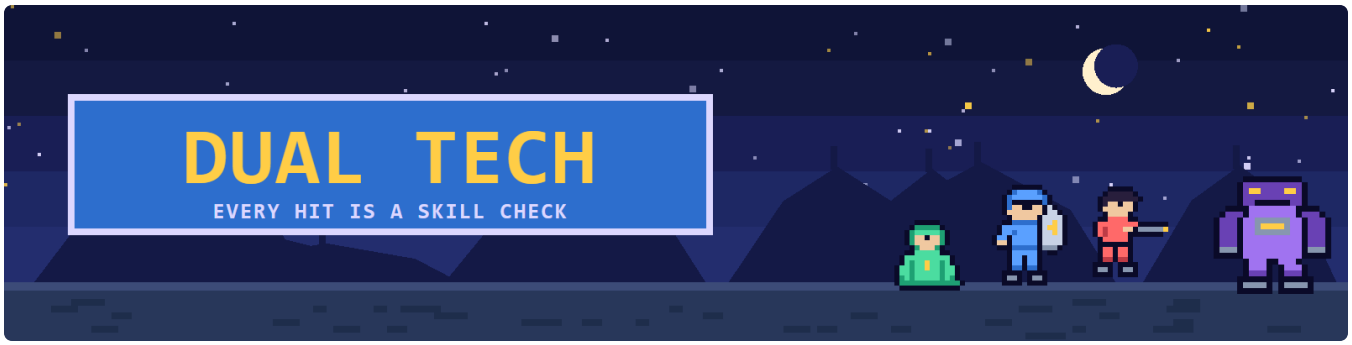
Dual Tech: game design



Oct 20, 2026 · @kr0

Status: draft. Rules marked "proposed" still need your decision; see the Proposed rules to confirm section in Open questions.

Related: Pitch · TDD · Open questions · Playable prototype

Changelog

Date	Change	Why
Oct 20, 2026	Polish pass: accessibility, screen flow and HUD, glossary, content counts, shield values, touch controls	Design critique, polish items
Oct 20, 2026	Added score, losing, grading rules, edge cases, tempo grid, upgrade draw rules, encounter fallback	Design critique, important items
Oct 20, 2026	Added timing windows, targeting, dual tech rules and formulas, enemy attacks	Design critique, critical items
Oct 20, 2026	Split into Pitch, GDD, TDD and Open questions; removed the pitch line, the one-month scope and the platform line	Owner's comments
Oct 20, 2026	First draft from the prototype fight	Prototype played and liked

Overview

Dual Tech is a short JRPG where every attack ends in a timing hit, and each hero's hit feels different. Brick holds and charges, Vex reacts, Lumen keeps the beat. When two heroes chain their hits, they fuse into a dual tech.

Design pillars

- **One hero, one feel.** Each hero's timing style is a different skill: patience (Brick), reflex (Vex), rhythm (Lumen). You should be able to tell who is acting with your eyes closed.
- **Decide, then perform.** The turn menu is the strategy, and the timing hit is a 1–2 second burst of skill. Neither half works alone.
- **Combos are the payoff.** Dual techs chain two feels back to back. Landing both perfectly is the best moment in the game.
- **Short and replayable.** One run takes 15–20 minutes, and you replay it to get more perfects and faster clears.

Content at a glance

Content	Count	Built in the prototype
Heroes	3	3
Hero abilities	6 (2 per hero)	6
Dual techs	6 (4 if the simultaneous ones are cut)	3
Encounters	5 (4 regular fights, 1 boss)	1 (the boss)
Enemy types	5	1
Upgrades	15 (12 hero, 3 Sync)	0
Music tracks	1 battle loop	0

Core loop

A turn takes about 5 seconds: wait, decide, perform, see the payoff. A run is five of those fights in a row, with an upgrade pick after each of the first four.

In a fight

1. Turn bars fill in real time. The enemy's bar fills too.

2. When a hero's bar is full, the fight pauses and you pick an ability or a dual tech.
3. You perform that hero's timing hit. The grade decides the effect.
4. Hits build Sync, which unlocks dual techs.
5. Repeat until the enemy or the party falls.

In a run

1. Fight one of the five encounters.
2. Pick 1 of 3 upgrades (after fights 1–4 only).
3. Move to the next fight. HP carries over and Sync resets.
4. Beat the boss to finish the run. You get a score from perfects, time and damage taken.

Losing (proposed). If every hero is down, the run ends. You see the results screen with your score so far, and a retry starts again from fight 1 with no upgrades.

Score (proposed). Shown at the end of every run, won or lost, with the best score saved.

```
\text{score} = 100 \times \text{perfects} + 2 \times \max(0,\ 1200 - \text{seconds}) - 3 \times \text{damage taken}
```

The score can't go below 0. Perfects count every graded hit and every rhythm beat. Seconds count only time when the clock is running, so time spent choosing commands is free.

Combat systems

The numbers below come from the prototype fight and are the starting point for tuning.

Turn bars (wait mode). Each hero's bar fills at its own speed. The fight pauses while you pick a command and during timing hits. The enemy only acts while time is running.

Timing grades

Grade	Damage multiplier	Sync gained
Perfect	1.5x (Railshot 2x)	+12
Good	1x	+5
Miss	0.4x	0

Rhythm hits grade each beat separately: +4 Sync for a perfect beat, +2 for a good one. The overall grade is Perfect if every beat is Perfect, Good if the beat score is at least half the number of beats, and Miss otherwise (proposed). Flurry grades each sweep separately, and its overall grade is its lowest sweep. Effects that check the grade, like stagger, use the overall grade.

Hit quality. Every graded hit or beat also has a quality score: Perfect 1, Good 0.6, Miss 0. Formulas below use it. Beat score is the sum of a rhythm hit's beat qualities, so 3 beats score 0–3.

Timing windows (proposed, taken from the prototype)

Style	Setup	Perfect	Good	Fails when
Hold (Brick)	Full charge in 1,250 ms; the zone is centred at 80% charge (1,000 ms)	±38 ms (±3% of the bar)	±125 ms (±10%)	Holding to 100% overcharges = Miss. No press within 3.5 s = Miss
Sweep (Vex)	The marker crosses the bar in 640 ms (Railshot), or 820 / 650 / 510 ms (Flurry's three sweeps). The zone centre is random, between 55% and 87% of the bar	±3% of the bar (±19 ms at 640 ms)	±9% (±58 ms at 640 ms)	Four crossings with no press = Miss
Rhythm (Lumen)	Beats 520 ms apart; first beat 1,000 ms after the hit starts	±70 ms	±140 ms	A beat passing 140 ms late = Miss. Pressing more than 140 ms but less than 320 ms early = Miss on that beat

Zone modifiers (proposed): Amp multiplies both window widths by 1.6. Hair trigger multiplies the Perfect window by 1.3. Modifiers multiply together, capped at 2x the base window.

Sync. A shared meter from 0 to 100. A dual tech costs 50 Sync. The partner must also have at least half a turn bar, and it loses that half.

Amp. Lumen's Amp field makes the zones on the next command 1.6x wider. It's the safety net for a hard dual tech or a player who is struggling.

Enemy pressure. Enemies telegraph through their own turn bar. Bosses get a second phase at 50% HP: faster turn bar and 1.35x damage.

Targeting (proposed)

- **Default target:** the enemy you hit last. At the start of a fight it's the leftmost enemy.
- **Changing target:** press left/right or tap an enemy while the command menu is open. The target can't change once the timing hit starts.
- **Single-target hits** (Shield slam, Railshot, Ricochet slam) hit the current target.
- **Multi-hit attacks** (Flurry's sweeps): each hit goes to the current target. If it dies, the remaining hits move to the next living enemy, left to right.
- **Area hits:** Prism shot hits every enemy. With more than one enemy alive, each one takes 60% of the damage.
- **Heals and shields** cover every living hero. Downed heroes only come back through a revive effect.
- **Taunt:** while a taunt is active, every enemy attack targets Brick if Brick is standing.

Edge cases (proposed)

- **Two bars fill on the same frame:** heroes act before enemies. Among heroes the order is Brick, Vex, Lumen; among enemies it's left to right. The others stay full and act next.
- **Downed heroes:** their bar stays empty and doesn't fill. They can't be a dual tech partner and can't be targeted. A revived hero comes back with an empty bar.
- **Last hit wins:** if the enemy's HP reaches 0 during a command, the fight ends right after that command resolves, even if a heal or a later hit would have followed.
- **Shields** stack, up to 40 per hero. They last until they absorb damage or the fight ends.

Heroes

Each hero is built around one timing style. Their ability, art, sound and combo role all come from that style.

Hero	Role	Timing style	Skill it tests	HP	Bar speed
Brick	Tank	Hold and charge	Patience, letting go at the right moment	140	Slow (0.20/s)
Vex	Damage	Reflex sweep	Reaction, stopping a moving marker	90	Fast (0.28/s)

Hero	Role	Timing style	Skill it tests	HP	Bar speed
Lumen	Support	Pulse rhythm	Rhythm, tapping on the beat	100	Medium (0.24/s)

Brick: hold and charge. You hold to fill a charge bar and let go inside the zone. Holding too long overcharges and misses. It should feel heavy and slow to build, with a big release.

- **Shield slam:** 34 base damage. A perfect hit staggers the enemy and resets its turn bar.
- **Barrier wall:** taunts the next 2 enemy attacks (Brick takes 30% less from them) and shields every living hero for 8 / 18 / 24 (Miss / Good / Perfect).

Vex: reflex sweep. A marker sweeps back and forth over a random zone, and you hit to stop it. Later sweeps get faster. It should feel sharp and twitchy, like landing a headshot.

- **Railshot:** one fast sweep for 48 base damage. A perfect hit does double.
- **Flurry:** three sweeps, each faster than the last, for 22 base damage per hit.

Lumen: pulse rhythm. Beats travel toward a ring at a steady tempo (520 ms apart), and you tap as each one crosses. It should feel calm and musical, a breather between the other two.

- **Pulse heal:** three beats. Heals every living hero by 6 + 10 x beat score (6–36). Getting every beat perfect revives downed allies at 30% HP.
- **Amp field:** one beat. Widens the zones on the next command and gives up to 20 Sync.

Dual techs

Most dual techs chain two heroes' timing styles back to back, with no pause between them. Holy charge and Overclock volley, if they ship, run both inputs at once. The switch between feels is the challenge. Each pair gets two: the prototype has the first three, and the other three are planned.

Starting a dual tech (proposed)

- Any hero with a full bar can open a dual tech from their command menu. They need a partner who is standing, has at least half a turn bar, and 50 Sync in the meter.
- Whoever opens it, the inputs always run in the order the table gives.
- Cost: 50 Sync. The opener's bar drops to 0. The partner's bar drops by half (a full bar drops to 50%).
- Sync earned by the dual tech's own hits counts. Sync is capped at 100, and any extra is lost.

Multipliers. "Mult" is the standard damage multiplier for the hit's grade: 1.5 / 1 / 0.4. Railshot's 2x applies only to Railshot, never inside a dual tech.

Dual tech	Pair	Input chain	Effect (proposed formula)	All-perfect bonus	Status
Ricochet slam	Brick + Vex	Hold-release, then a sweep (600 ms)	60 x (Brick mult + Vex mult) damage to the target (max 180)	x1.5 damage (270) and the enemy is staggered	In prototype
Bulwark hymn	Brick + Lumen	3 beats, then hold-release	Each living hero heals 8 + 8 x beat score (8–32). Each living hero gets a shield of 10 + 20 x hold quality (Miss 10, Good 22, Perfect 30). Taunt for 1 attack	Revives downed heroes at 40% HP	In prototype
Prism shot	Vex + Lumen	2 beats, then a sweep (620 ms)	40 x (1 + beat score) x Vex mult to every enemy (max 180; 60% each with 2 or more enemies)	x1.3 damage and 30 Sync refunded	In prototype
Breach	Brick + Vex	Shield slam, then 3 Flurry sweeps	Flurry sweeps can't grade below Good: 22 x mult per sweep	Every hit on the next hero turn counts as a crit (x1.5)	Planned, values TBD
Holy charge	Brick + Lumen	Tap beats while holding; each beat adds charge	Stuns every enemy; the length scales with beats landed	Stun lasts 2 enemy turns	Planned, pending cut #1
Overclock volley	Vex + Lumen	Stop the marker on each of 4 beats	4 shots, 18 x mult each	Both heroes' bars refill	Planned, pending cut #1

Holy charge and Overclock volley run two inputs at the same time instead of one after the other. They are the hardest to build and tune, so they go first on the cut list. If they're cut, Brick + Lumen and Vex + Lumen have one dual tech each.

Enemies and encounters

Each fight puts one hero's timing style in the spotlight, so by the boss you have practised all three. The HP values are first guesses, and the scrap golem's 700 HP comes from the prototype.

#	Encounter	HP	Bar speed	Attack (proposed)	Targets	Twist	Spotlight
1	Rust rats (x3)	60 each	0.30/s each	Bite: 8	A random hero	Weak and fast; a tutorial fight	Vex: Flurry clears the group
2	Bell warden	300	0.18/s	Toll: 14	The lowest-HP hero	Every 3rd action it rings the bell instead: beats switch to double-time (260 ms apart) for the next 2 hero turns	Lumen: rhythm under pressure
3	Plated beetle	400	0.16/s	Ram: 20	A random hero	Armor: Good and Miss hits do 0 damage; only Perfect hits get through	Brick: Shield slam to stagger, then Vex
4	Twin drones	250 each	0.22/s each	Zap: 10	A random hero	On its turn, a drone that hasn't been hit since its last turn heals its partner 25 instead of attacking	Dual techs: Prism shot hits both
5	Scrap golem (boss)	700	0.17/s, 0.25/s after overheat	Slam: 16–26 (x1.35 after overheat)	A random hero	Overheats at 50% HP	Everything: taunt, heal and big dual techs

Enemy rules (proposed)

- An enemy acts when its bar is full. It picks its target at that moment, using the Targets column. Taunt overrides this.
- Damage hits shields first, then HP. A taunted attack on Brick does 30% less damage.
- Stagger resets an enemy's bar to 0. An enemy can't be staggered again until it has acted once, which prevents stun-locking the boss.
- With several enemies, each has its own bar, and they act in the order their bars fill.

The golem is the boss in the prototype. Overheat at 50% HP is its only phase change.

- **Laser phase** (stretch goal, cut list #2): a telegraphed laser that you block with Brick's hold. It adds a defensive input that appears nowhere else, so it depends on the open question about blocking enemy attacks with timing.
- **If encounters 2 and 4 are cut** (proposed fallback): the golem rings Bell warden's bell every 3rd action once it overheats. Lumen keeps a rhythm-under-pressure moment, and the golem fight becomes the dual tech test.

Progression

After each of the first four fights you pick 1 of 3 upgrades from a pool of 15, so a run has 4 picks and none after the boss (proposed). Upgrades change how a hero's timing feels, not just their numbers, so each run leans into different styles.

- **Brick:** Steady grip (charge bar fills 20% slower) · Aftershock (a perfect slam also shields Brick for 15) · Iron will (overcharging counts as Good instead of Miss; it doesn't help against the Plated beetle, whose armor ignores Good hits) · Wide stance (Barrier wall taunts 3 attacks)
- **Vex:** Hair trigger (perfect zone 30% wider, sweep 15% faster) · Fourth shot (Flurry gets a 4th sweep) · Hollow point (Good hits on Railshot do 1.4x) · Ricochet (a missed sweep counts as Good)
- **Lumen:** Encore (Pulse heal gets a 4th beat) · Slow tempo (Lumen's own beats switch to half-time, 1,040 ms apart, so they stay on the music grid) · Resonance (each perfect beat gives +2 extra Sync) · Lingering amp (Amp lasts 2 commands)
- **Sync and dual techs** (new, proposed): Quick sync (dual techs cost 40 Sync) · Tag team (the partner needs only a quarter bar) · Harmony (an all-perfect dual tech also gives +20 Sync)

Draw rules (proposed): the 3 offers never include an upgrade you already have, and they can't all belong to the same hero. If fewer than 3 are left, show what's left.

Permanent unlocks between runs are out of scope for the first release.

Controls, feel, art and audio

One button plays every timing hit, so the challenge is the timing, not the controls.

Controls. Keys 1–4 or a click pick a command. Space, a tap or the Hit button plays the timing hit (press for Vex and Lumen, hold and release for Brick). The same input works on keyboard, mouse and touch. On touch (proposed): commands are four large buttons along

the bottom of the screen, you tap an enemy to target it, and during a timing hit the whole lower half of the screen is the hit button. The final layout depends on which platforms ship first (see Open questions).

Feedback for each hero

Hero	On screen	Sound	Perfect hit
Brick	Charge bar fills, screen tightens	Low rising hum, heavy thud on release	Hit-stop for 120 ms, big screen shake
Vex	Thin bright marker, fast motion	Sharp click on each sweep	Crack sound, short white flash
Lumen	Beats pulse toward a ring	Soft chime on the tempo	Chord that builds with each perfect beat

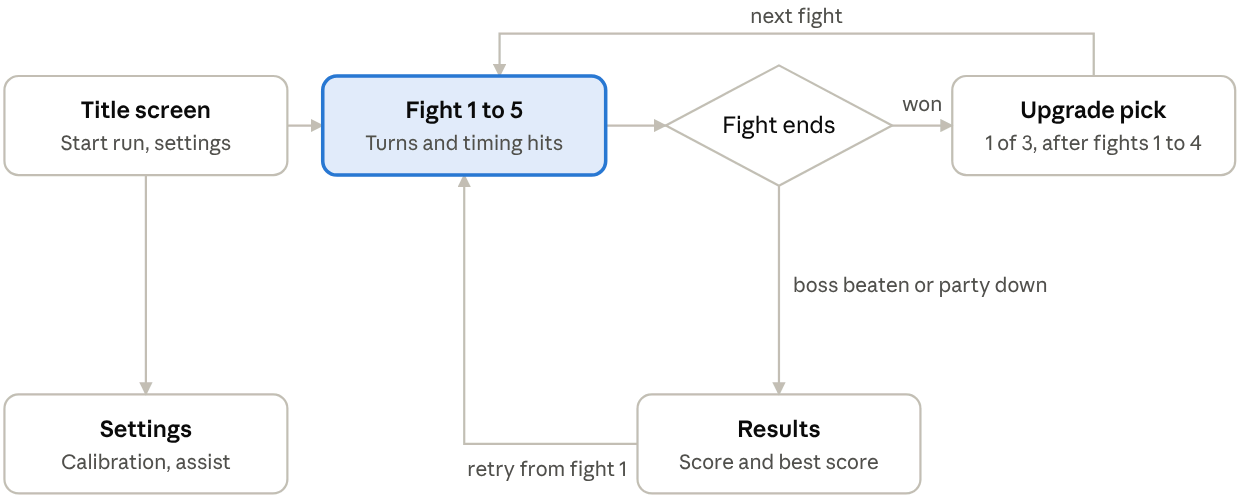
Dual techs get a name banner, a slight slow-motion moment between the two hits, and a combined sound.

Art. Chunky pixel sprites in side view, like the prototype: night sky, blue Chrono Trigger-style menu windows, and one colour per hero (blue Brick, red Vex, green Lumen). Use that colour on the hero's sprite, health bar and timing bar so you always know who is acting.

Audio. One looping battle track at a fixed tempo of about 115 BPM (one beat is 522 ms, matching Lumen's 520 ms spacing). Lumen's beats always land on the music grid, so the music doubles as the rhythm guide. Anything that changes beat spacing snaps to a subdivision of the track: Bell warden switches to double-time and Slow tempo to half-time (proposed).

Screens and HUD

A run loops fight and upgrade until the boss, then results and retry



screen flow · 5 screens, 1 loop

A run is one loop: fight, then pick an upgrade, four times, then the boss. Winning or losing goes to the results screen, and a retry starts again from fight 1.

HUD during a fight (proposed)

Element	Shows	Where
Hero panel	Name, HP, shield, turn bar and hero shape for each hero; the acting hero is highlighted	Bottom left
Enemy bars	HP and turn bar above each enemy; the current target is marked	Above each enemy
Command menu	The acting hero's 4 commands, with the reason a dual tech is unavailable	Bottom right
Sync meter	0–100, highlighted at 50 or more	Under the hero panel
Timing meter	The active hit's bar, zones and step count ("Hit 2 of 3")	Centre bottom, only during a hit
Grade pop-up	Perfect / Good / Miss, damage and heal numbers	Over the hero or enemy

Element	Shows	Where
Status icons	Taunt, stagger, Amp, overheat	Beside the affected hero or enemy
Run tracker	Fight number (1–5) and the run timer	Top

Accessibility

Every timing cue reaches at least two senses, and every timing window can be loosened. All of these are proposed.

- **Colour:** red Vex and green Lumen are hard to tell apart with red-green colour blindness. Each hero also gets a shape on their bars and grade pop-ups (Brick a square, Vex a diamond, Lumen a circle), and the Good and Perfect zones get different patterns, not just different colours.
- **Sound and sight:** every audio cue has a visual one. The "eyes closed" pillar is a goal for sound design, not a requirement to play by ear.
- **Motion:** separate toggles for screen shake, hit flashes and slow motion.
- **Timing assist:** a setting that widens every window by 1.5x (it stacks with Amp, inside the 2x cap). Scores from runs with assist on are marked. See the easy-mode question in Open questions.
- **Latency calibration:** a player-facing screen in settings: tap along to 8 beats, and the game stores your offset. The engine side is in the TDD.
- **Hold alternative:** an option to play Brick's hold as press to start and press again to release, for players who can't hold a button.
- **Text:** grade words (Perfect / Good / Miss) always appear beside the colour and sound feedback.

Glossary

Term	Meaning
Amp	Lumen's buff: the next command's timing windows are 1.6x wider

Term	Meaning
Beat score	The sum of a rhythm hit's beat qualities (Perfect 1, Good 0.6, Miss 0)
Dual tech	Two heroes chaining their timing hits into one move; costs 50 Sync
Grade	The result of a timing hit: Perfect, Good or Miss
Mult	The damage multiplier for a grade: 1.5 / 1 / 0.4
Overcharge	Holding Brick's charge to 100%; counts as a Miss
Overheat	The boss's change at 50% HP: a faster bar and 1.35x damage
Quality	A grade as a number: Perfect 1, Good 0.6, Miss 0
Stagger	Resets an enemy's turn bar to 0; can't repeat until the enemy has acted
Sync	The shared 0–100 meter, filled by good hits and spent on dual techs
Taunt	Forces enemy attacks onto Brick, who takes 30% less from them
Turn bar	Fills over time; when it's full, that hero or enemy acts
Wait mode	The clock stops while you choose a command or perform a timing hit